

Title xxxxxxxxxxxxxxxxxxxxxxxxxxxx

Abstract

of the system.

1. Introduction

Smart

integration.

2. Literature review

Multimodal

Table 2: Comparison between traditional, unimodal, and the proposed multimodal identification systems

3. Methodology

3.1 Overview

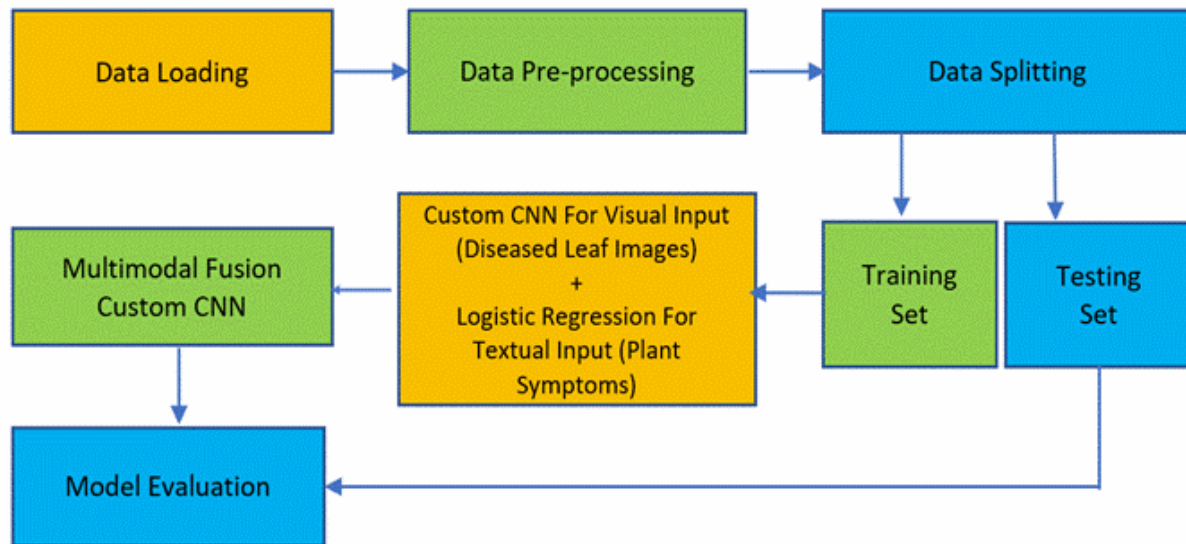


Figure 1: Proposed Methodology

3.2 Data Acquisition

In

3.3 Dataset Description

Textual

actual on-field images without the support of high-tech cameras. There are 1162 images of healthy corn leaves, 1306 images of corn leaves with Common Rust, 1146 images of corn leaves with Blight, and 574 images of corn leaves with Gray Leaf Spot. The dataset is under-sampled to 500 samples per class.

Figure 2: Visual Dataset a. Healthy, b. Common Rust, c. Blight, d. Gray Leaf Spot

Table 3 shows sample records of the textual dataset. The fields are showing symptoms of three corn diseases, namely Common Rust, Blight, and Gray Leaf Spot. A single, simply worded statement is added as symptoms as described by layman farmers for practical use.

Table 3: Textual Datasets Sample

Disease	Symptom 1	Symptom 2	Symptom 3
Common Rust	Small, reddish-brown pustules	Yellow halos around pustules	Leaves are starting to curl and become brittle
Blight	Oblong lesions tapering at the ends	Dead leaf tissue between veins	Tan lesions on leaves
Gray Leaf Spot	Gray or tan rectangular lesions	Lesions running parallel to leaf veins	Chlorotic streaks

3.4 Preprocessing

Text preprocessing

3.5 Model Architecture

Figure 3 disease classes [33].

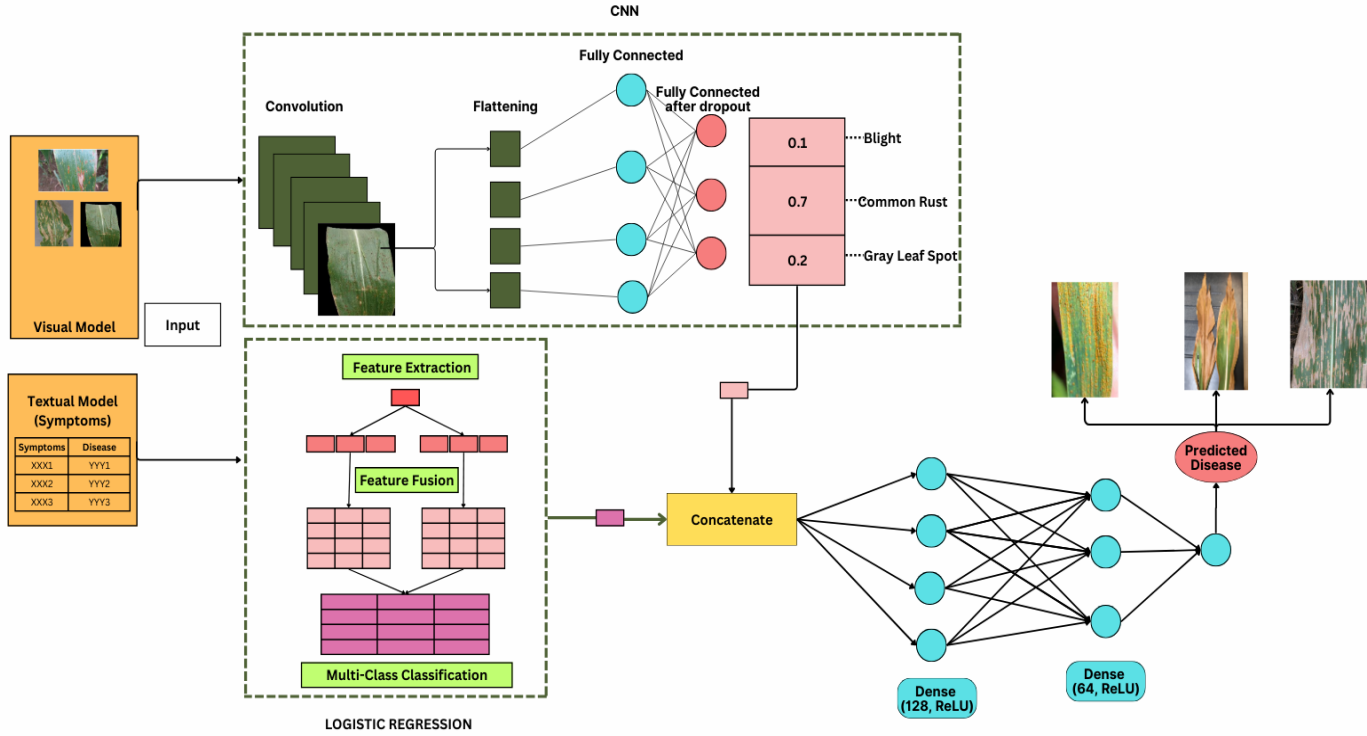


Figure 3: Model Architecture of the proposed work.

Linear Regression Equation:

$$z = w \cdot X + b \quad (1)$$

Sigmoid Equation:

$$\sigma(z) = \frac{1}{1+e^{-z}} \quad (2)$$

Logistic Regression Equation:

$$P(X; b, w) = \frac{1}{1+e^{-(w \cdot X + b)}} \quad (3)$$

filters with ReLU a

based on their relative importance. A Logistic Regression classifier is used to pair symptoms with diseases based on feature vectors generated from TF-IDF vectorization.

For multimodal integration, the image feature vector generated by CNN and the text feature vector are combined to generate a unified feature vector for multimodal fusion. The unified feature vector is

then passed to a where the availability of substantial paired image and text data corpora is not guaranteed.

4. Results

The proposed

Software Environment: Python 3.10, TensorFlow 2.13 / PyTorch 2.0, CUDA 11.8, Jupyter Notebook / VS Code/ Colab.

4.1 Performance Metrics and Performance Evaluation

After

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Figure 4: Training, Validation Loss, and Accuracy of the proposed model

Figure 5: Precision-Recall Curve

Evaluation of ROC graph performance was performed under four class categories, namely: Blight, Common Rust, Healthy, and Gray rate, reflecting the model's excellent ability to distinguish between diseased and healthy samples.

Figure 6: ROC Curve

Figure 7 presents the t-SNE plot used for visualising high-dimensional feature space. It shows well-separated clusters for each class, confirming that the model has learned distinct and meaningful feature representations that allow effective discrimination between disease categories and healthy leaves.

Figure 7: t-SNE plot

Figure 8 depicts the confusion matrix, where the model has high classification accuracy across all four classes with very few misclassifications which indicates strong precision and recall balance along with high F1-score.

Figure 8: Confusion Matrix

As reflected in the confusion matrix, the majority of misclassifications occur between Blight and Gray Leaf Spot, which exhibit visually similar symptoms such as overlapping lesion shapes, spot density, and texture patterns. This symmetric confusion indicates inherent inter-class similarity rather than model instability. Common Rust remains mostly distinguishable due to its characteristic red-brown rust pustules, while the Healthy class shows minimal confusion. The limited and structured nature of these errors suggests that misclassifications arise from inherent inter-class similarity rather than model instability, confirming the effectiveness of the multimodal fusion framework.

In table 4, there is an ablation study results comparing different model configurations based on the modalities used, fusion strategy, and classification accuracy. This analysis highlights the individual contributions of image and textual modalities and demonstrates that combining both significantly improves performance, with the proposed CNN + LR model achieving the highest accuracy.

Table 4: Ablation Report of the proposed model with unimodal methods

Model Configuration	Modalities Used	Fusion Strategy	Accuracy (%)
CNN-only	Image	NA	97.02%
Logistic Regression	Text	NA	92.13%
Proposed (CNN + LR)	Image + Text	Late fusion	98.85

4.2 Comparative Analysis

Table 5

Table 5: Comparison of the proposed model with existing works

Ref	Main Techniques	Accuracy	Important Features
[40]	Vision Transformer (ViT) + CNN	78.00%	The paper represents a multimodal approach integrating image and text data using cross-attention mechanisms to classify cassava leaf diseases.

[41]	SVM + CNN	94.84%	This paper uses small sample data by integrating image, text, and labelled data. It uses vision-language pre-training and contrastive learning. The method performs well even with limited data and effectively generalizes to other scenarios.
[42]	DenseNetDNN	96.1 %	It only works on the image dataset of Corn leaves.
Our Work	Logistic Regression + CNN	98.85%	Multimodal approach to identify Corn leaf diseases.

5. Conclusion

In this

References